

Lecture 26: Application of the NP lemma

— Construction of MP (size- α) tests

Two examples — one in each category to distinguish between the non-randomized and randomized tests.

Consider a random sample $x_1, \dots, x_n$ from $\mathcal{S}_\theta$, a continuous prob. distn.
By the NP lemma, the MP (size α) critical region is for $H_0: \theta = \theta_0$ ag. $H_A: \theta = \theta_1$ (any θ_1)

is, $w_0: \frac{f_{H_A}(x)}{f_{H_0}(x)} \geq k \Leftrightarrow T(x) \geq k'$. So the test fn is

$$\phi(x) = \begin{cases} 1 & \text{if } T(x) \geq k' \\ 0 & \text{if } T(x) < k' \end{cases}$$

where k' is $\ni E_{H_0} \phi(x) = P_{H_0}(T(x) \geq k') = \alpha$ either 0 or 1.

This is a non-randomized test, ϕ takes only 2 values, 0 or 1.

Ex $x_1, \dots, x_n$ r.s. from $N(\mu, \sigma^2)$, σ^2 known.
 $H_0: \mu = \mu_0$ ag. $H_A: \mu = \mu_1 (> \mu_0)$

c.r. $w_0: \frac{f_{H_A}(x)}{f_{H_0}(x)} \geq k \Leftrightarrow \exp. \frac{1}{2\sigma^2} [\sum (x_i - \mu_0)^2 - \sum (x_i - \mu_1)^2] \geq k$

$$\Leftrightarrow 2(\mu_1 - \mu_0) \sum x_i - n(\mu_1^2 - \mu_0^2) \geq k'$$

$$\Leftrightarrow \sum x_i \geq k'' \quad k'' = \frac{k' + n(\mu_1^2 - \mu_0^2)}{2(\mu_1 - \mu_0)} \quad \therefore \mu_1 > \mu_0.$$

So, the test fn is $\phi(x) = \begin{cases} 1 & \text{if } \sum x_i \geq k'' \\ 0 & \text{if } \sum x_i < k'' \end{cases}$, k'' from the size condn

ie, $P_{H_0}(\sum x_i \geq k'') = \alpha \Rightarrow \Phi\left(\frac{k'' - n\mu_0}{\sigma\sqrt{n}}\right) = 1 - \alpha = \Phi(\tau_\alpha) \Rightarrow k'' = \sigma\sqrt{n}\tau_\alpha + n\mu_0$

ϕ , thus specified, is a non-randomized test.
What is the Power of this test?

Power of the test = $P_{H_A}(\sum x_i \geq \sigma\sqrt{n}\tau_\alpha + n\mu_0) = 1 - \Phi\left(\tau_\alpha - \frac{\sqrt{n}(\mu_1 - \mu_0)}{\sigma}\right)$

which increases in the sample size n
the diff. betn $(\mu_1 - \mu_0)$

decreases in σ , the variability in the popn

with logical implications, but all of these are usually fixed (known, or unknown) (σ, μ_0, μ_1) (τ)

but, the power also increases with decrease in τ_α , ie, increases in α (here, the prob. of type I error)

This makes perfect sense, $\therefore$ power = 1 - Prob. of type II error,
 $\star \exists$ a trade-off between the two probs!

In case of discrete distributions, the exact size- α condition^{*} may not be achievable $\forall \alpha \in (0,1)$. This necessitates the use of a randomized test.

$x_1, \dots, x_n$ r.v.s. from to (discrete). For testing $H_0: \theta = \theta_0$ ag. $H_A: \theta = \theta_1$ ($\theta_1 > \theta_0$)

The MP (size- α) test, like, non-randomized test can be constructed from the

NP lemma as,

$$\phi(x) = \begin{cases} 1 & \text{if } T(x) > c \\ \gamma & = \\ 0 & < \end{cases}$$

c & γ are obtained from (exact) size- α condition, i.e., $E_{\theta_0} \phi(x) = \alpha$ or $\sup_{\theta \leq \theta_0} E_{\theta} \phi(x) = \alpha$.

Ex.

X is a single obsn from $\text{Bin}(n, \theta)$. To test $H_0: \theta = \theta_0$ ag. $H_A: \theta = \theta_1$ ($\theta_1 > \theta_0$) at size α .

By NP lemma, c.r. of the MP size- α test is

$$w_0: \frac{f_{H_A}(x)}{f_{H_0}(x)} \geq k \Leftrightarrow \left\{ \frac{\theta_1(1-\theta_0)}{\theta_0(1-\theta_1)} \right\}^x \left(\frac{1-\theta_1}{1-\theta_0} \right)^n \geq k \Leftrightarrow x \geq k^* \quad \left(\begin{array}{l} \text{The ineq. } \geq \text{ stands} \\ \because \theta_0, \theta_1 \in (0,1) \\ \text{and } \theta_0 < \theta_1 \end{array} \right)$$

k is obtained from the size- α condition, i.e., $P_{H_0}(X \geq k^*) \leq \alpha$. The ~~non~~ non-randomized test in that case is $\phi(x) = \begin{cases} 1 & \text{if } x \geq k^* \\ 0 & x < k^* \end{cases}$.

where $k^* \ni E_{\theta_0}(\phi(x)) \leq \alpha$

But if ~~it is~~ the exact size- α condition needs to be satisfied, then we have to consider the randomized test. k^* can be obtained using size- α condition, i.e., $P_{H_0}(X \geq k^*) \leq \alpha$, and

$$\phi(x) = \begin{cases} 1 & \text{if } x > k^* \\ \gamma & = \\ 0 & < \end{cases}$$

γ from the exact size- α condition, i.e., $E_{\theta_0} \phi(x) = P_{H_0}(X > k^*) + \gamma P_{H_0}(X = k^*) = \alpha$. $\hookrightarrow (\alpha)$

In this example, if $P_{H_0}(X \geq k_1) > \alpha$ and $P_{H_0}(X \geq k_2) \leq \alpha$ for the first time, then $k^* = k_2$.

i.e., k^* is the min $k \ni P_{H_0}(X \geq k) \leq \alpha$, i.e., ~~if $P_{H_0}(X \geq k_1) > \alpha$~~ $P(x > k_1) > \alpha$ and $P(x > k_2) < \alpha$, with $k_2 = k_1 + 1$ and, $P(x > k_1) = \alpha_1$ and $P(x > k_1 + 1) = P(x > k_2) = \alpha_2$, then

$$(*) \Rightarrow \phi(x) = \begin{cases} 1 & \text{if } x > k_2 \\ \gamma & = \\ 0 & < \end{cases} \quad \text{where } \gamma = \frac{\alpha - \alpha_2}{\alpha_1 - \alpha_2} (> 0)$$

if $\alpha_2 = \alpha$, the non-randomized test will suffice.



but there is no such pt. in $\mathbb{R}$.

Thus, if $x = k_2$ is actually obsd in the sample, we need to construct a test which would reject $H_0: \theta = \theta_0$ w.p γ .